

1. Sum and difference formula of tan?

tan :

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} \text{ (Handouts)}$$

$$\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

2. Sum and Difference Formula of Tan And Sin?

sin :

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A-B) = \sin A \cos B - \cos A \sin B$$

tan :

(Handouts)

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

3. Pythagorean Identities? 3 Mark

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta \text{ (Handouts)}$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

4. Trigonometric Identities? 2 Marks

Basic Trigonometric Identities:

Reciprocal Identities:

$$\csc \theta = \frac{1}{\sin \theta}, \sec \theta = \frac{1}{\cos \theta}, \cot \theta = \frac{1}{\tan \theta} \text{ (Handouts)}$$

Quotient Identities:

$$\tan \theta = \frac{\sin \theta}{\cos \theta}, \cot \theta = \frac{\cos \theta}{\sin \theta}$$

5. Find Exact Value Of 210? 5 Marks

$$\cos(210^\circ) = \cos(180^\circ + 30^\circ)$$

$$= \cos 180^\circ \cos 30^\circ - \sin 180^\circ \sin 30^\circ$$

$$= (-1) \left(\frac{\sqrt{3}}{2} \right) - (0) \left(\frac{1}{2} \right)$$

$$= -\frac{\sqrt{3}}{2}$$

(Practice Solve 25 to 23 page 2)

6. Define descriptive statistics and inferential statistics?

7. Cumulative frequency?

8. Coefficient of variation?

9. Perpendicular line?

10. Pie chart?

11. Trigonometry property?
12. Geometric progression and Arithmetic progression?
13. Find Exact Value Of $\cos 150^\circ$
14. Find Exact Value Of $\sin 75^\circ$
15. Solve $\sin \theta = k$
16. Relative Frequency of F 15, 12, 10, 28
17. Find Angles For Urdu English Pie Chart,
18. Find Class Width Of F 12 & F Density 0.6,
19. Find Equation Of Line Passing From Points 6, 7 & Is Perp To $2x + 3y = 8$
20. Find 50% Percentile?
21. Find 25% Percentile?
22. Find Line Of Equation If Point 7, 7 Is Slope Of $x + 3y = 8$?
23. Find Median?
24. Find 1st Term And Common Difference If 2nd Term Is 13 And 7th 68
25. Find A³⁷
26. If $R = 1.06$ And $A = 300000$
27. Find The Line Of Equation If Point (7, 7) Is The Slope Of $x - 3y = 8$
28. 38 42 49 25 31 67 61 57 53 49 42 36 29 Find The Five Number Summary
29. Find Harmonic Mean
30. Find Central Point And Radius
31. Find Slope (2, 1), (3, 1)
32. Find Quartile Equation
33. Find Arithmetic Mean
34. Find Standard Deviation
35. $1 + \tan \theta / 1 + \cot \theta = \tan \theta$
36. Find the slope of (2, 6), (4, 8)
37. $\tan \theta - 1 = k$
38. Iqr (inter-quartile range)
39. Compute c
40. 5 number system
41. Slope
42. $\sin 1/2$ So $\sin \theta$ - $180^\circ < \theta < 180^\circ$ Find All $\sin$ Value
43. Find Slope Perpendicular, Parallel of $2x + 3y = 21$ and $3x - 2y = 18$? 3 Marks
44. Find Frequency If Frequency Density Is $= 0.6$ And Class Width Is 12? 3 Marks
45. Find Standard Deviation Of 35, 42, 47, 29, 41, 45, 52, 37, 59, 62, 27? 5 Marks
46. Find the arithmetic mean of following data? 10, 20, 23, 20, 23, 27, 24, 23, 29
47. $\sin 120^\circ$ Value?
48. $\cos 120^\circ$ Value?
49. Verify the following equation $1 + \tan \theta / 1 + \cot \theta = \tan \theta$?
50. Find the first two term in the expression of $(4 - x)$ in ascending power of x ?
51. Write the step to find $\tan \theta - k$?
52. Expand the difference formula $\sin (A - B)$?
53. Find inter quartile range when third value ie is 33 and First quart ie is 23?
54. Find the slope of line through the point (3, -1) & (4, -2)